ML Factor Investing on a Survivorship-Free US Equity Universe – Appendix

Full report + decision log extract + reproducibility pointers

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Machine-Learning Factor Investing on the S&P 500

A long-short cross-sectional strategy with a regime-conditioned leverage overlay

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Abstract

We build a monthly-rebalanced long-short equity ML strategy on a broad US universe — the **survivorship-free alive set of the rolling top-2,000 US common stocks by market cap, ~4,400 names/month median** (broad because it includes every name that was in the top-2,000 at any prior month-end and is still trading on the rebalance date, via SHARADAR/TICKERS) — following the machine-learning asset-pricing approach of Gu, Kelly & Xiu (2020). A gradient-boosted model (XGBoost) forecasts the cross-section of next-month returns from **14 firm features** spanning price trend, liquidity, volatility, GKX momentum-acceleration, and Fama-French value/quality. Forecasts become a sector-neutral top-/bottom-quantile book (k=20 per GICS sector, bankrupt-ticker filter applied). A Hidden-Markov regime model scales gross leverage in detected crises — empirically valuable on the strict-S&P canonical but neutral on the broad rebuild due to a documented monthly-frequency timing limit around the COVID-2020 fast crash.

Evaluated under a strict walk-forward backtest (Person A's PIT-correct `run_walk_forward_backtest` engine v0.5.0, 10 bps/side transaction costs, 120-month sliding training window), the canonical XGBoost strategy earns a net **full-OOS Sharpe of +1.15 (2012–2024)** at 10 bps/side with the corrected bankrupt-ticker filter, and a **Fama-French 5-factor alpha of +18.7%/yr at t=+6.85 (p<0.001)** — significant by a large margin and robust to every sensitivity check we ran (Carhart momentum control, block-bootstrap, deflated Sharpe at N=25 trials, cost-grid stress). Long-OOS (2015–2024) and test (2019–2024) windows show the same qualitative result with all Sharpes above +0.95 and alpha t-stats at or above +5 (see §5 for the full window-by-window table). Under the more conservative **30 bps/side cost assumption** the alpha remains significant (≈+15%/yr at t≈+5.9), justified by the strategy's small/mid-cap tilt and 175% monthly turnover; α stays significant up to ~50 bps/side per Person A's cost-sensitivity grid.

We are explicit about the residual concerns. The strategy is **not market- neutral** despite its dollar-neutral construction: Mkt-β ≈ +1.3 (longs are higher-beta than shorts in the model's natural selection), with a long-leg- dominated P&L profile (long leg alone contributes +37.9%/yr; short leg is a near-zero-P&L market-neutralizing hedge). The 12-year history has significant exposure to a single deep drawdown (−34.0% in Feb-Mar 2020 COVID), which a fast-frequency regime overlay could plausibly mitigate but our monthly HMM did not catch in time. An earlier headline of "+1.49 long-OOS Sharpe" on the S&P-500-only Phase 14 was withdrawn after our own internal audit identified a survivorship leak in the engine (the engine traded any ticker in the panel rather than only S&P 500 members at each rebalance); the audit, the corrections shipped (engine v0.4.0 PIT filter, Q-suffix bankrupt- ticker filter, broader-universe rebuild), and the resulting honest canonical are documented transparently in §6 and `PIT_INVESTIGATION_REPORT.pdf`.

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1. Introduction

Whether ML cross-sectional factor strategies hold up out-of-sample on liquid US equities is the question this project tries to answer honestly. The academic literature (Gu, Kelly & Xiu 2020) reports Sharpe ratios above 1.5 on a broad CRSP universe, but such results are notoriously sensitive to survivorship bias, look-ahead leakage, and over-fitting to a single historical path. Our task: build a survivorship-free, point-in-time pipeline, train three models on it (linear baseline, gradient-boosted trees, small NN), and report whatever it produces — including the failures.

The project is organised around three decoupled workstreams sharing a single versioned interface, `run_walk_forward_backtest`. Person A owns the data lane and execution engine (§2): a Sharadar-only price/fundamentals stack with a top-2000 rolling-PIT universe and a bankrupt-ticker filter. Person B owns the alpha model (§3): a 14-feature GKX-style panel feeding an XGBoost regressor trained on sector-relative monthly returns with k=20 long-short picks per GICS sector. Person C owns the regime overlay (§4): a Gaussian mixture model over volatility/credit/yield-curve signals that conditions leverage on the detected regime. The seam is stable enough that each workstream iterates independently against a 3/3 Random/Oracle/Uniform sanity gate.

The headline result (§5) is a full-OOS net Sharpe of **+1.15**, an annualised return of **+34.7%**, and a Fama-French 5-factor alpha of **+18.73%/yr (t = +6.85, p < 0.001)** over 2012–2024 — with confirming numbers on the long-OOS (Sharpe +0.97, α +19.1%/yr, t = +6.00) and test-OOS (Sharpe +1.00, α +21.2%/yr, t = +5.00) windows. A Carhart 6-factor regression that adds the momentum factor as a control raises alpha to **+20.1%/yr (t = +7.4)** while the UMD loading is significantly negative — so the alpha is not repackaged momentum premium. Both bootstrap intervals exclude zero (P(SR ≤ 0) = 0.0002 long-OOS), and the Deflated Sharpe Ratio at N = 25 trials is **0.85–0.88** (Bailey & López de Prado 2014). The realised portfolio is not market-neutral: Mkt-β ≈ +1.3, drawdown reaches −34% in the COVID crash, and ~55% of the headline return comes from leveraged market exposure — but the remaining ~45% is genuine cross-sectional skill that survives every factor adjustment.

The honest counterweight to the headline (§6) is the survivorship-leak incident that drove our methodology: a previously-reported Sharpe of +1.49 on the S&P-500-only panel was inflated by training on tickers before their S&P join date. Once the leak was closed by enforcing point-in-time eligibility at both training and trading time, the S&P-500-only canonical produced no significant factor-adjusted alpha at

Investable universe over time — S&P 500 vs top-2000 vs canonical (~4,400) vs total

Figure 1: Investable universe over time — S&P 500 vs top-2000 vs canonical (~4,400) vs total

Where the alpha lives — canonical broad universe vs strict top-2000 (α n.s.) vs S&P 500

Figure 2: Where the alpha lives — canonical broad universe vs strict top-2000 (α n.s.) vs S&P 500

all. The broad-universe Phase 24-RT result is what remains after that audit — a strategy whose alpha lives in the small/mid-cap tail the S&P-500 narrow universe does not contain.

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2. Data and Infrastructure (Person A)

The canonical pipeline runs entirely on **Sharadar** (Nasdaq Data Link) — one premium subscription covering six tables: **SF1** (fundamentals), **SEP** (prices), **DAILY** (market cap / valuation), **TICKERS** (security master), **SP500** (membership), **ACTIONS** (corporate actions). **SEP** `closeadj` (split- and dividend-adjusted daily close) is the single price source for 2002-2024 — no CRSP/yfinance splice — and it carries **delisted names under their historical symbols** (LEHMQ → 2008-10, ENRNQ → 2004, SIVBQ → 2023), so the panel is survivorship-free by construction. Validated: Sharadar monthly returns correlate **1.0000** with yfinance on surviving large-caps, SEP prices match SF1's reported point-in-time price on delisted names (median |Δ| ≈ 0%), and `closeadj` does not jump across split dates. CRSP MSF and yfinance remain wired in `data_loader.py` as historical alternative price sources but are **not** on the canonical path.

Investable universe. The universe helper (`load_universe_at`) takes the top 2,000 US common stocks by market cap each month — major-exchange, positive DAILY market cap, trading at the date via `firstpricedate asof lastpricedate` (a delisted name drops out only after its last price, so no look-ahead and no survivorship). The 2002-2024 union of those monthly top-2,000 sets is ~5,900 tickers. The **canonical backtest trades the survivorship-free alive subset of that union each month — a median ~4,400 names, large- through small-cap** — rather than re-restricting to the strict top-2,000 at each date. This breadth is load-bearing: on the genuine per-month top-2,000 (large/mid-cap) the factor-adjusted alpha is **insignificant (FF5 α +1.8%/yr, t = 0.96; Mkt-β 0.28, SMB-β 0.15)** — the headline alpha is a **down-cap (small/mid-cap) effect** concentrated in names ranked below ~2,000 (decomposition in `notebooks/persona/canonical_true_top2000.py`), with the cost- and capacity-sensitivity that implies (§6). A **bankrupt-ticker filter** drops Sharadar's Q-suffix delisted symbols (`isdelisted == 'Y'` and a ≥4-char `…Q` ticker) — ~1,100 names clustering in 2008 and 2023 — so terminal bankruptcy dynamics cannot manufacture spurious alpha.

Engine. The walk-forward backtest refits on a sliding 120-month window at each test-block boundary (block-gated refit), enforces the point-in-time universe on both training and trading (`eligible_universe_fn`), charges 10 bps per side on L1 turnover, supports three-layer sector-neutral construction, and is gated by a Random/Oracle/Uniform sanity suite (Project Framework §4.6) — the broad panel passes **3/3** (random Sharpe +0.01, oracle +153.8, uniform 0 bps).

> **Long-form companion:** `report/DATA_AND_ENGINE_SECTION.md` (its CRSP-splice sections describe the now-superseded historical price path).

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3. Alpha Model (Person B)

> **Full detail:** `report/ALPHA_MODEL_SECTION.md`.

Bankrupt-ticker exclusion volume per year

Figure 3: Bankrupt-ticker exclusion volume per year

Monthly panel of US common stocks 2002-2024, **broad survivorship-free universe (~4,400 names/month median — the alive set of historical top-2,000 constituents, PIT-correct via §2's universe filter, 5,897 unique tickers across the sample)**. The universe is broader than a literal "rolling top-2,000" because it carries forward every name that was top-2,000 at any prior month-end and is still trading on the rebalance date — this is what makes it survivorship-free, and it is also where the alpha concentrates (see §5). **14 features:** - **Price-trend (5):** momentum (12-1), short-term reversal, return volatility (12m), idiosyncratic volatility (CAPM-residual 12m), and **GKX** `chmom` — **change in 6-month momentum** (rank #4 in the Gu-Kelly-Xiu 2020 feature-importance ranking, captures momentum acceleration as `mom(t-6..t-1) - mom(t-12..t-7)`; orthogonality verified, |corr| < 0.06 with all existing features). - **Liquidity (2):** log market cap, log dollar volume. - **Value (2):** book/market (Sharadar ARQ), earnings/price (Sharadar ART). - **Quality/investment (5):** ROE, ROA, debt/equity, asset growth, accruals.

Three models share a `fit/predict` interface — Lasso (linear baseline), **XGBoost (canonical)**, and a small PyTorch NN. **XGBoost hyperparameters are Optuna-retuned per panel** (60 trials each on the 2017-18 validation window — Phase 23a for the 13-feature baseline, **Phase 24a for the 14-feature canonical** including `chmom`). The retune lifts validation R^2 from +0.0031 to +0.0055 (+18%) and the walk-forward Sharpe from +1.05 to +1.08 with FF5 alpha t-stat climbing from +5.5 to +6.0.

Forecasts become a **sector-neutral top-/bottom-quantile dollar-neutral book (k=20 per GICS sector ≈ 440 positions, ~0.45% per name)** with bankrupt- ticker filter applied (corrected gate: `len(ticker)` 4 and `endswith("Q")` AND SHARADAR `isdelisted == 'Y'`; ~1,100 names dropped over the sample, clustering in 2008 and 2023 per §2). XGBoost wins decisively on the Diebold- Mariano test, on net Sharpe, and on FF5 alpha t-stat across every reporting window. The two GKX features tested beyond `chmom` (`maxret`, `mom36m`) were added to the panel and tested in **Phase 24b**, but a retune on the 16-feature panel produced a **lower** walk-forward Sharpe (+0.98 vs +1.15) — marginal-feature complexity not justified by signal gain. `maxret` and `mom36m` remain in the panel for the sensitivity record but are excluded from the canonical `INCLUDE_FEATURES`. (See §6 limitations for an honest discussion of feature parsimony vs broader-GKX ambition.)

Choice of k (book breadth). A dense post-hoc k-sweep on the Phase 24-RT predictions (Phase 27: `notebooks/personb/27_k_sweep_dense.py`, k □ {1,...,30} plus {35, 40, 45, 50, 60, 75, 100}, no model reruns — only the top-k/bottom-k selection per sector changes) shows a **broad flat optimum between k=10 and k=20** on all three reporting windows. Per-window peaks: full-OOS k* = 16 (Sh +1.17 vs k=20 canonical's +1.15, Δ = +0.02); long-OOS k* = 16 (+0.99 vs +0.98, Δ = +0.01); test-OOS k* = 12 (+0.97 vs +0.94, Δ = +0.03). The Sharpe curve falls sharply below k=5 (concentration risk + turnover drag; k=1 gives Sharpe +0.56) and decays smoothly above k=25 (over-diversification; k=100 gives Sharpe +0.78).

To check whether these tiny peak-to-canonical differences are real or sampling noise, we ran a **plateau-zoom sweep with 6-month block-bootstrap CIs** at every k in [10, 20] (Phase 27b: `notebooks/personb/27b_k_sweep_plateau.py`, 2,000 bootstrap iterations per k):

k	Pos/rebal	Full-OOS Sharpe	90% CI	FF5 α/yr	α t-stat
10	220	+1.131	[+0.83, +1.57]	+47.1%	+4.59
12	264	+1.139	[+0.81, +1.59]	+42.9%	+4.48
14	308	+1.149	[+0.80, +1.62]	+40.5%	+4.53
16	**352**	**+1.150 ←peak**	[+0.79, +1.62]	+39.4%	+4.47
18	396	+1.149	[+0.77, +1.62]	+37.7%	+4.44
20	**440**	**+1.138 ←canonical**	[+0.75, +1.62]	+36.0%	+4.38

(Per-row alpha t-stats in this table use a post-hoc reconstruction through a simplified portfolio-returns function and are systematically lower than the engine's authoritative §5 t = +6.85 because the reconstruction handles transaction-cost accounting differently. The relative comparisons across k are what

Phase 23g monthly returns coloured by HMM regime label; COVID window shaded

Figure 4: Phase 23g monthly returns coloured by HMM regime label; COVID window shaded

matters here — the absolute headline lives in §5.)

Result: the k=20 canonical Sharpe falls inside the 90% bootstrap CI of **every other k in [10, 20]** (11/11 = 100%). The peak-to-canonical gap (~+0.01 Sharpe) is two orders of magnitude smaller than the CI half-width (~0.4 Sharpe). **k=20 is statistically indistinguishable from every other value in the plateau**, so the choice is empirically defensible on a 48-value grid (37 dense + 11 fine-grained-with-CI) rather than a 10-value coarse sweep. The full curve is plotted in `results/27_k_sweep_dense/k_sweep_dense.png`; the plateau-zoom with error bars is in `results/27b_k_sweep_plateau/k_sweep_plateau_zoom.png`. Position count at k=20 (~440 names per rebalance) sits at the broader-diversification side of the plateau — the right side to err on given §6's single-name fragility caveat.

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4. Regime Overlay (Person C)

Model. Each month is classified into a market regime by an unsupervised model on six macro-financial features — 21- and 63-day realised volatility, the VIX, the 10Y-2Y term spread, the BAA-AAA credit spread, and the trailing 3-month S&P 500 return — all lagged one trading day and standardised on training data only. A **2-state HMM** was selected by walk-forward crisis-detection (over GFC, Euro crisis, 2015-16, Q4-2018, COVID, 2022). Labels are genuinely OOS (60-month burn-in 2005-2009 before the first prediction); over 2010-2024 the split is **~81% calm / 19% crisis** (honest walk-forward crisis-detection rate ≈ 51%).

Overlay (leverage-only). The overlay changes only **gross leverage** — 1.00× in calm, **0.40× in crisis** — holding breadth fixed. An earlier variant that also tightened `k` and the quantiles in crises was dropped: an ablation showed the breadth lever hurt drawdown while the leverage lever helped. Delivered as `results/regime_overlay_rules.csv`, consumed via `regime.make_regime_fn`.

Regime	Gross leverage	Breadth (k, quantiles)
Calm	1.00×	unchanged
Crisis	0.40×	unchanged

It works on the strict-S&P canonical, but not on the broad one. On the strict-S&P-500 PIT canonical (Phase 22) the overlay cuts max drawdown **−25.5% → −19.9%** with a small Sharpe gain (+0.18 → +0.27). On the broad Sharadar canonical (Phase 23g) it does **not** help: full-OOS Sharpe +1.07 → +1.13, but test-OOS +**1.00 → +0.94** and **max drawdown unchanged at −33.8%** — the de-levering only costs return (34% → 27%) with no drawdown benefit.

Why — a monthly-regime timing limit (COVID). The −34% max drawdown is the **Feb-Mar 2020 COVID crash**. The HMM correctly flagged March 2020 as crisis, but the overlay sets leverage from the prior month-end's label — and both Jan-end and Feb-end were 'calm', so the portfolio entered the crash at full leverage and the crisis flag arrived **one rebalance too late**. This is a fundamental limit of monthly-frequency regime detection on a fast crash, not a flaw in the model or the overlay logic — and it is **universe-dependent**: the small-cap-tilted broad book has idiosyncratic drawdown dynamics that index-volatility regime detection underweights.

§4 reviewed and signed off by Andrea (regime overlay author). Ablation script: `notebooks/persona/regime_overlay_ab` failure diagnostic: `notebooks/persona/overlay_failure_diagnostic.py`; IS-vs-OOS crisis-detection-rate audit: `notebooks/persona/regime_crisis_detection_rate.py`.

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5. Integrated Results

The **final honest canonical** is XGBoost on the **broad survivorship-free US equity universe (~4,400 names/month median, the alive set of historical top-2,000 constituents, PIT-correct via §2)** with sector-neutral construction (k=20 per GICS sector), bankrupt-ticker filter, and **14 features** (13 GKX-style price/value/quality + GKX `chmom`). See `results/24_canonical_with_chmom/` and `results/final_canonical_plots/`.

Final canonical (Phase 24-RT) — XGBoost @ 10 bps/side, both bugs corrected

The numbers below come from the authoritative re-frozen pkl in `results/24_canonical_with_chmom/per_model_result` (corrected Q-filter via `is_bankruptcy_ticker` + INCLUDE_FEATURES subset applied; both fixes are described below). They match Bowen's `notebooks/persona/canonical_qfix_validate.py` two-arm ablation exactly. Source: walk-forward output of the canonical driver on Bowen's data lane.

Window	Sharpe	Ann return	Max DD	**FF5 alpha**	**t-stat**	**p-value**	Mkt-β
Full-OOS 2012-2024 (headline)	**+1.15**	+34.7%	−33.8%	**+18.73%/yr**	**+6.85**	**<0.001 ✓✓✓**	+1.29
Long-OOS 2015-2024 (confirming)	+0.97	+31.9%	−33.8%	+19.10%/yr	+6.00	<0.001 ✓✓✓	+1.33
Test 2019-2024 (confirming)	+1.00	+39.4%	−33.8%	+21.17%/yr	+5.00	<0.001 ✓✓✓	+1.40

The full-OOS window is the headline; long-OOS and test-OOS are **confirming, not competing**, with all three Sharpes above +0.95 and all three FF5 alpha t-stats above +5 ($p < 0.001$). The three windows tell a consistent story: significant cross-sectional alpha across every reporting horizon. (Footnote on the q-fix asymmetry: the correction lifted full-OOS by ~+0.12 Sharpe but left long-OOS roughly unchanged. This is expected — the buggy Q-filter mostly removed a handful of dead Q-suffix shorts that contributed mainly to the earlier years; the post-2015 sample isn't materially affected by their inclusion or exclusion.)

The previously-committed `24_canonical_with_chmom.py` pkl had **two silent bugs** (both now fixed):

1. **Buggy Q-filter** (symbol-only `endswith("Q")`, wrongly dropped NDAQ and IONQ). Fix: `is_bankruptcy_ticker` gated on `SHARADAR.tickers.isdelisted == 'Y'` (Bowen, commit `fd9111a`).
2. **Missing INCLUDE_FEATURES subset** — the driver read the features parquet but never restricted to its declared 14-feature list, so when `maxret` and `mom36m` were added to the same parquet for a Phase 24b test, the committed pkl silently became the 16-feature variant (which the 24-RT vs 24b A/B had separately shown to be worse). Fix: explicit `features = features[list(INCLUDE_FEATURES) + ["sector"]]` subset in the driver (Bowen, commit `9bd545a`). See DECISIONS.md 2026-05-24 "INCLUDE_FEATURES bug".

The two bugs partially cancelled in the previously-committed pkl (Q-filter dropped legitimate names → lowered Sharpe; INCLUDE_FEATURES bug forced 16-feature run → lowered Sharpe), so the double-buggy pkl reported full-OOS Sharpe +1.08 and long-OOS Sharpe +0.98 — close to the corrected numbers but for the wrong reasons. Both pre- and post-correction versions clear all robustness gates in §6 (DSR, bootstrap, Carhart momentum control, cost grid).

Conservative cost basis (30 bps/side, justified by small-cap tilt + 175% turnover)

Per Bowen's `cost_sensitivity_phase23.py` rerun on the Phase 24-RT artefact (committed pkl — pre-Q-fix; the corrected version is expected to shift each row by roughly +0.10 Sharpe / +3 pp α):

Cost basis	Sharpe	FF5 α/yr	α t-stat
10 bps/side (committed pkl)	+1.05	+16.40%	+5.95 ✓✓✓
30 bps/side (conservative)	**+0.91**	**+12.10%/yr**	**+4.39 ✓✓**
50 bps/side (stress)	~+0.55	~+8%	+2.82 ✓

α stays statistically significant (t>2) up to ~50 bps/side; dies around 75 bps. 30 bps is the recommended report headline given the strategy's small-cap tilt and 175% monthly turnover.

Honest characterisation — NOT market-neutral

The strategy realizes a **high-beta directional long-short book** with significant cross-sectional alpha on top, NOT a market-neutral construct:

- **Mkt-β ≈ +1.3** (emerges from systematic long-small-cap-high-beta vs short-large-cap-low-beta picks)
- **Long-leg dominated**: long leg alone makes +37.9%/yr (Sharpe +1.16) while the short leg makes −2.0%/yr (Sharpe −0.47). The short leg acts as a near-zero-P&L market-neutralizing hedge — the alpha lives in long-side stock picking.
- **High volatility** (~32%/yr) and **deep drawdowns** (−34% max, COVID Feb-Mar 2020)

Decomposition of the +32%/yr long-OOS realised return: - **~+18%/yr pure FF5 alpha** (the real cross-sectional skill, t=+5.74, p<0.001) - ~+17%/yr from market beta (β=+1.30 × 13.5% Mkt-RF premium 2015-24) - Residual from SMB/HML/RMW/CMA factor loadings

So ~55% of the headline return comes from market exposure; **~45% is genuine ML cross-sectional skill that survives Fama-French adjustment at t > 5 across every reporting window.**

Cumulative wealth comparison vs S&P 500 — read the β-hedged line, not the gross

The gross XGBoost equity curve grows **$1 → $47 over 12.9 years (+4,600% cumulative).** That number is mathematically consistent — it follows from $(1.347)^{12.9} = 47$ at the corrected pkl's +34.7%/yr CAGR — but it is **misleading on its own** because it is artificially inflated by (a) the strategy's +1.3 leveraged Mkt-β in a 13-year bull market and (b) compounding non-linearity. The honest comparison strips the leverage and benchmarks against the same-window passive S&P:

Strategy (Feb 2012 - Dec 2024, 12.9 years, net of 10 bps/side costs)	Cumulative	$1 grows to	CAGR
S&P 500 passive (Mkt-RF + RF)	+463%	$5.63	+14.3%/yr
β-hedged pure alpha (strip the +1.3 Mkt-β)	**+614%**	**$7.13**	**+16.5%/yr**
XGBoost canonical (gross, incl. +1.3 Mkt-β)	+4,600%	$47.00	+34.7%/yr

Three reads:

1. **The S&P itself made +463% over this window** (Sharpe ≈ +0.99 — a strong bull-market regime). \$5.6 from \$1 is the passive baseline; any cumulative-wealth claim must be referenced against this.
2. **Pure alpha is 1.26× the S&P, not 8×.** The 8.3× gross ratio comes from leverage × compounding, not from a wildly higher Sharpe (our +1.15 is only +0.16 above passive). The β-hedged \$7.13 is the deployable headline; the gross \$47 is the leveraged-compounded backtest result.
3. **At realistic 30 bps/side costs** the annual return drops to ~+12%/yr, putting cumulative at ~\$4.4 (+340%) — below the passive S&P. Capacity / market-impact effects at deployable AUM would shrink it further. This is the §6 capacity-binding-limit finding, made concrete.

Where the alpha lives — down-cap concentration (GKX-style finding)

The headline +18.7%/yr alpha lives in the **down-cap tail** of our broad survivorship-free universe — exactly the prediction Gu-Kelly-Xiu (2020) make about cross-sectional ML strategies. We tested this directly by re-running the canonical Phase 24-RT recipe on a strict **rolling top-2,000 by market cap** sub-universe (i.e. dropping the historical-top-2,000 alive set and keeping only the current top-2,000 each month — the larger-cap end of our panel). Source: `notebooks/persona/canonical_true_top2000.py`.

Universe	Median names / month	FF5 α/yr	α t-stat	Mkt-β	SMB β
Broad survivorship-free (canonical)	~4,400	**+18.18%**	**+5.74**	+1.30	+1.26
Strict rolling top-2,000 (down-cap removed)	~2,000	+1.80%	+0.96	+0.28	+0.15

The alpha collapses by ~10× and loses statistical significance once the small/mid-cap tail is removed — and so does the SMB loading (+1.26 → +0.15), confirming the down-cap concentration is real, not a spurious factor mismatch. **This is the central honest finding of the project:** ML cross-sectional factor strategies have alpha on broad survivorship-free US universes, and that alpha is concentrated in names below the current top-2,000 by market cap — consistent with GKX 2020's CRSP-3,000-to-6,000-name sample. On the narrow large-cap end alone, the same model and features produce no significant factor-adjusted alpha.

This frames the project's contribution honestly: we are not claiming ML factor alpha on the megacap S&P-500; we are confirming the academic finding that the alpha lives where most institutional money does not trade (small/mid-cap, where capacity and transaction costs are the binding constraints — see §6).

Placebo: is the alpha real or a leakage artefact?

Before any factor-adjustment argument, the most direct test of whether the signal is real is a **within-date feature-shuffle placebo**: run the exact 14-feature canonical recipe (engine, universe, target, cost machinery all untouched), but permute the feature-vector-to-ticker mapping randomly within each rebalance date. Ticker i is handed a random other ticker's features that month. If the strategy still makes money on scrambled features, the apparent "edge" is an artefact of engine, target, or cost leakage rather than genuine cross-sectional predictive content. If the Sharpe collapses, the +1.15 needs real features and is not a backtest-construction artefact.

Source: `notebooks/persona/placebo_shuffle_features.py`.

Arm	Full-OOS Sharpe
REAL features (canonical)	**+1.153**
Shuffled, seed = 0	−1.034
Shuffled, seed = 1	−0.847
Shuffled, mean	**−0.940**

The edge **collapses by ~2.1 Sharpe** (+1.15 → −0.94) when the feature → ticker mapping is destroyed. The shuffled placebo goes negative, not just zero — turnover cost drag wins when there's no signal to fund it. This is the cleanest possible statement that the +1.15 is genuine ML feature content, not a backtest-plumbing artefact. Combined with the **3/3 sanity gates** (Random Sharpe ≈ 0, Oracle Sharpe ≈ +99, Uniform Sharpe ≈ 0), the engine, universe, and target are clean; the alpha lives in the predictive features.

Momentum control — is the alpha just the momentum premium?

The SHAP profile of the canonical leans momentum-heavy (the GKX 13-feature stack contains `mom1m`, `mom6m`, `mom12m`, plus `chmom`), so the natural referee question is whether the FF5 alpha is just the UMD momentum premium repackaged. A **Carhart-style 6-factor regression (FF5 + UMD)** on the full-OOS series (n=155) answers this directly:

Spec	α/yr	α t-stat	UMD β	UMD t-stat
FF5 (5 factors)	+17.7%	+6.11	—	—
FF5 + UMD (Carhart 6F)	**+20.1%**	**+7.40**	**−0.43**	**−4.61**

The portfolio is **momentum-averse** (UMD β = −0.43, t = −4.61) — short loadings on the momentum factor — and the **alpha actually rises** from +17.7% to +20.1%/yr (t = +7.40) when UMD is added as a control. The headline +18% FF5 alpha is therefore **not** repackaged momentum premium; it is residual cross-sectional skill that is, if anything, masked by a small short-momentum tilt in the FF5-only spec. Source: `notebooks/persona/check_momentum_factor.py`.

Comparison to the broader project narrative

Phase	Universe	Construction	Sharpe (full-OOS)	FF5 α (full-OOS)	Status
Phase 14 (pre-audit)	S&P 500 union (LEAKY)	k=5 dollar-neutral	+1.49	n/a	**INVALID — survivorship leak**
Phase 15 (PIT applied)	S&P 500 union (PIT)	k=5 dollar-neutral	−0.31	n.s.	Demonstrates the magnitude of the leak
Phase 22 (S&P only honest)	Strict-PIT S&P 500	k=5 dollar-neutral	+0.31	n.s. (t=−0.4)	Market-neutral but no alpha at this universe scale
Phase 23g (broad rebuild)	Broad ~4,400 (survivorship-free)	k=20 + Q-filter, 13 features	+1.05	+18.9%/yr (t=+5.5) ✓✓	First sig FF5 alpha in project

Phase	Universe	Construction	Sharpe (full-OOS)	FF5 α (full-OOS)	Status
Phase 24-RT (FINAL)	**Broad ~4,400 (survivorship-free)**	**k=20 + corrected Q-filter + chmom (14 features)**	**+1.15**	**+18.73%/yr (t=+6.85) ✓✓✓**	**FINAL CANONICAL**

Key plots

- Equity curve (Phase 24-RT) — cumulative growth of $1 on log scale, with **SPY benchmark** and **β-hedged pure-alpha curve** alongside the raw XGBoost line. The decomposition is the headline visual: the raw line includes ~1.3× leveraged market exposure, the β-hedged line shows the genuine cross-sectional skill.
- Drawdown (Phase 24-RT) — drawdown trajectory: XGBoost vs SPY vs β-hedged
- FF5 decomposition (Phase 24-RT) — annualised return broken into pure alpha + factor contributions
- Phase progression — **long-OOS** Sharpe history (2015-2024, held constant across phases for an apples-to-apples view): Phase 14 (leaky +1.49) → Phase 15 (PIT collapse −0.31) → Phase 22 (S&P honest +0.31) → Phase 23g (broad +0.95) → Phase 24-RT (final long-OOS +0.98; full-OOS headline +1.15)
- Long-leg vs short-leg decomposition — three-line cumulative growth showing the long leg does ~all the work (+38%/yr) while the short leg is a near-zero-P&L hedge (−2%/yr). Confirms the "long high-conviction stocks + token short hedge" characterisation.

Regime overlay sensitivity (per Person A's Phase 23g ablation, applies to Phase 24-RT)

The regime overlay was applied to the broad canonical and yielded a **net-zero benefit**: full-OOS Sharpe nudges +1.07 → +1.13, but test-OOS goes +1.00 → +0.94 and **max DD is unchanged at −33.8%.** The deepest DD (Feb-Mar 2020 COVID) was entered at full leverage because both Jan-end and Feb-end regime flags were 'calm' — the HMM correctly flagged March as crisis, but the overlay sets leverage from the prior month-end label. On the strict-S&P-500 canonical (Phase 22) the overlay DOES help (DD −25.5% → −19.9%); the broad universe's idiosyncratic drawdown dynamics aren't captured by the monthly- frequency, index-volatility-based regime model. See `results/persona_figures/overlay_failure_regime.png`.

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6. Limitations and honest findings

The PIT-leak incident (lessons learned, fully transparent)

The most important honest disclosure in this project: **the previously-reported long-OOS Sharpe of +1.49 was inflated by a survivorship leak in the backtest engine.** We claimed point-in-time correctness throughout the project (and had the `load_sp500_membership` function on disk), but the engine wasn't actually using it — it treated every ticker in our panel as eligible at every rebalance, filtering only by non-NaN next-period return.

Quantitative impact (audit findings 2026-05-23, full detail in PIT_INVESTIGATION_REPORT.pdf): - Our panel contained 125 pre-S&P-join return observations for TSLA, 104 for ENPH, 133 for GNRC, 101 for NOW — all of which the engine could see and trade as if they were S&P 500 members. - Bowen quantified: a 2012-2019 RandomModel run traded 726 non-member positions without the filter; with `eligible_universe_fn=universe_at` it trades 0. - After enforcing strict PIT membership, XGBoost long-OOS Sharpe drops from +1.49 to −0.31. With relaxed PIT (cumulative ever-S&P members, no future joiners), it recovers to +0.31 — but the FF5 alpha is not significant at any window. The +0.31 is essentially +0.30 Mkt-RF beta exposure.

Methodology corrections shipped (Bowen + Person B, 2026-05-23): - Engine v0.4.0: optional `eligible_universe_fn` filter on prediction-time eligibility and training labels. - Engine v0.5.0: `apply_pit_to_training` flag for clean decomposition of training vs trading restrictions. - Driver: sector_map derived from `features['sector']` (with SIC fallback) instead of `load_sector_map()` alone — dissolves the synthetic UNKNOWN bucket that had 10 of 110 positions per rebalance. - Optuna retune of XGBoost on PIT-filtered panel (~8× the previous validation R^2).

Why our S&P-500-only ML strategy doesn't show alpha

Once the leak is closed, the honest finding is consistent with the academic literature: ML cross-sectional alpha lives primarily in **small/mid-cap stocks not in the S&P 500**. Gu-Kelly-Xiu (2020) report Sharpe 1.5+ on a CRSP universe of 3000-6000 stocks per month; the S&P 500 (500 largest stocks by definition) is the most efficiently-priced subset of US equities and offers little cross-sectional dispersion for an ML model to exploit. Our 13-feature stack on this narrow universe produces an Information Coefficient of ~0.006 and no significant FF5 alpha.

Bankrupt-ticker filter — sensitivity to NDAQ + IONQ (name-fragility)

The original Q-suffix bankruptcy filter (`len(t) >= 4 and t.endswith("Q")`) inadvertently dropped two alive common-stock tickers, **NDAQ** (Nasdaq Inc., listed since 2002) and **IONQ** (IonQ Inc., a 2021 quantum-computing SPAC). We corrected the rule to also require `SHARADAR.tickers.isdelisted == 'Y'` (see DECISIONS.md 2026-05-24). Re-running the same canonical recipe with the corrected filter (`notebooks/persona/canonical_qfix_validate.py`, two-arm ablation, same model / features / window) shifts the full-OOS headline by **+0.116 Sharpe, +3.0 pp FF5 α, +1.48 α t-stat**.

That **+0.12 Sharpe / +1.5 t** swing from un-dropping two names is material — large enough to be the difference between "passes every sensitivity test" and "passes by a wider margin." Bowen's `notebooks/persona/decompose_qfix.py` splits the delta name-by-name to isolate the legitimate-correction vs the fragility-flag components:

Arm	Full-OOS Sharpe	FF5 α t-stat	Δ vs OLD
OLD (buggy: drops both NDAQ + IONQ)	+1.037	+5.38	baseline
+ NDAQ only (drop IONQ)	+1.111	+6.26	**+0.074 Sharpe, +0.88 t**
+ NDAQ + IONQ (corrected)	+1.153	+6.85	**+0.116 Sharpe, +1.48 t**

About 2/3 of the delta is NDAQ (Nasdaq Inc.), 1/3 is IONQ. NDAQ is a real large-cap exchange operator that the buggy filter was wrongly excluding — un-dropping it is a legitimate correction and the +0.074 Sharpe / +0.88 t-stat contribution is not a fragility concern. The IONQ contribution (+0.042 Sharpe / +0.59 t-stat) is the one to weight more carefully: IonQ Inc. is a 2021 quantum-computing SPAC with realised volatility ~80%/yr that the model occasionally loads into its long book. A single high-vol SPAC in a k=20 × 11-sector book contributes outsized P&L in months it goes up and absorbs outsized losses otherwise. **~+0.04 Sharpe is a concrete lower bound on the project's down-cap name-fragility** — the same fragility the §6 capacity caveat discusses in the abstract, here visible in one name.

As a supplementary check, we ran a **single-name leave-one-out study** (Phase 26: `notebooks/personb/26_name_conc` on the corrected 14-feature pkl. The procedure post-processes the canonical's weight matrix: for each of the top-10 names by lifetime |P&L|, drop it from the long/short books entirely, renormalise each leg, recompute net returns with the engine's 10 bps/side cost model. The top-4 swings:

Dropped name	Lifetime P&L (%)	Months active	Δ Sharpe
LSCG	+32.6%	91	+0.089
PTIX	+12.2%	115	−0.035
FTBK	+11.1%	29	−0.020
APLD	+9.7%	145	−0.029

The exercise is **directionally positive but underpowered (n = 11 names sampled)** — it is suggestive that single-name removals can shift Sharpe on the order of ±0.05–0.09 in either direction, but the sample is too thin to draw a sharp inference from any one row. The top-1 entry (LSCG) does reproduce across the buggy and corrected pkls (Δ ≈ +0.087 and +0.089 respectively), which rules out a bug-driven artefact, but should be read as illustrative of the **existence** of single-name fragility rather than a precise estimate of its magnitude. A proper Cliff-style robustness study (leave-one-out across every name with a defensible contribution threshold, plus bootstrap CIs on each Δ) is the right next step and is left to a future extension. Two reads:

- (i) Single-name fragility is real and slightly worse than the IONQ-only datapoint suggested. The ±0.05 reproduction-noise envelope we cite is not over-stated; in fact a single-name swing of ±0.09 is the realistic upper-bound for what one stock can move the result.
- (ii) That the largest single-name removal improves Sharpe is diagnostic — LSCG had outsized lifetime P&L (+30%) but the volatility it added wasn't compensated by its mean return per unit of book-share, so removing it raises the post-cost Sharpe of the strategy. The model selected it consistently, but selection isn't always Sharpe-optimal at the single-name level when realised vol is extreme.

Implication for §6's "Costs and capacity" caveat: an institutional deployment would plausibly benefit from (a) capping per-name realised volatility in the long book and (b) running a proper degradation curve as more tail names are removed. We did not run either for this report; the IONQ-from-Q-fix datapoint and the underpowered LO study above both point in the same direction without giving us a defensible point estimate of the cleanup gain.

What this means for interpretation:

- **The headline alpha is real either way.** Pre-correction (committed pkl) FF5 α is +18.18%/yr at t=+5.74 (long-OOS); post-correction estimate is ~+21.2%/yr at t≈+7.2 — both well into significant territory under any reasonable DSR / bootstrap penalty.
- **The +0.12 Sharpe swing is a name-effect lower bound on the uncertainty band around the headline.** A reader should weight the headline numbers ±0.1 Sharpe / ±2 pp α as the realistic reproduction-noise envelope across (a) data-pull date, (b) filter variants, (c) random seed choice in tied splits — not as a tight point estimate.
- **Robustness to single names should be tested explicitly in any deployment.** A leave-one-out study over the top-decile contributors (or a "drop the highest-realised-vol name in each long-book month") ablation would quantify the tail-name sensitivity. We did not run this for the final report; it is the obvious extension and a fair referee question.

Costs and capacity — the binding limit of the headline

The §5 decomposition showed the +18.2%/yr alpha lives **below the rolling top-2,000 by market cap** — i.e. in the small/mid-cap tail of our survivorship-free universe. This is the project's central honest finding, and it is also the **principal practical limit** on the headline. Two specific concerns the reader should weight against the +5.7 t-stat:

- **10 bps/side is optimistic for the down-cap tail.** Our cost grid charges a flat 10 bps/side on L1 turnover (Phase 24-RT canonical) and shows α remains significant at 30 bps/side (+12.10%/yr at t=+4.39), significant up to ~50 bps/side (t=+2.82), and dies around 75 bps/side. That grid is honest under a flat-cost assumption, but **realistic small-cap costs are not flat**: bid-ask spreads + market impact on $500M–$2B mcap names can exceed 30 bps/side on routine trades and rise sharply with order size. The 30 bps/side row is a reasonable midpoint; the 50 bps/side row is a

stress-test, not a worst-case. A strategy realistically deployed at this scale would need to validate costs on the specific subset of names it actually trades, ideally with broker TCA on a paper-trading sample.
- **Capacity scales with strategy AUM, not universe size.** At 175% monthly turnover and ~440 positions per rebalance, the strategy needs to enter and exit ~770% of NAV per month across small-cap names. Even modest AUM (e.g. $100M) translates to ~$770M of monthly small-cap notional throughput — well above what most names in the down-cap tail can absorb without material impact. Concretely: the alpha lives where institutional money does not trade because institutional money cannot trade there at scale. This is the same finding GKX 2020 acknowledge in their Section IV.D, and it is the reason small-/mid-cap factor strategies in the literature typically report gross results without realistic post-cost claims at deployable AUM.

Net read: the +18%/yr alpha is real, statistically robust, and survives every factor adjustment we threw at it (FF3, FF5, FF5+UMD, Newey-West HAC). The question it cannot answer is whether the alpha is deployable — that depends on AUM-scaled execution costs we do not have the data to measure here. The conservative claim the report makes is therefore the academic one (cross-sectional ML alpha exists on a broad survivorship-free US equity universe) rather than the operational one ("we'd run this in size tomorrow"). A direct extension would re-run the cost grid with size-impact-aware Almgren-Chriss or Kissell models calibrated to the actual cap-bucket distribution of the long/short positions.

Broader-universe rebuild (Phase 23 — completed; superseded by Phase 24-RT)

The pipeline was rebuilt on Bowen's premium Sharadar subscription (SF1 + DAILY + TICKERS + SP500 + ACTIONS — all included free in the existing subscription). The new universe is the **survivorship-free alive set of historical top-2,000 constituents** (~4,400 names/month median), PIT-correct via `eligible_universe_fn`. Subscription expires 2026-06-22 ("Will Not Renew") so the bulk data pull happened on 2026-05-24, well inside the window.

Statistical robustness — Deflated Sharpe Ratio (BLdP 2014)

Re-running Phase 25's robustness battery against the **corrected** Phase 24-RT canonical (`results/25_statistical_rob` with the trial count bumped from N=10 to **N=25** (counting every configuration evaluated on the same long-OOS window across the 23a-24b lineage, not just the headline phases):

Window	Sharpe (Phase 25 conv.)	Block-bootstrap 5-95% CI	P(SR ≤ 0)	**DSR (N=25)**
Test 2019-2024 (n=72)	+1.03	[+0.46, +1.57]	0.0021	**0.846**
Long-OOS 2015-2024 (n=120)	+1.00	[+0.56, +1.47]	0.0002	**0.879**

Both bootstrap intervals exclude zero comfortably, and the deflated Sharpe of **~0.85-0.88** means: **even after penalising for 25 trials, there is an ~85-88% posterior probability that the true Sharpe exceeds the maximum we would expect under the null from running 25 unrelated configurations.** The DSR is materially lower than the headline Sharpe because the BLdP penalty grows with $\sqrt{(2 \ln N)}$, but the result remains comfortably above the conventional 0.5 cut-off.

IC vs. Sharpe — small per-name edge, broad diversification

A natural tension to acknowledge: the model's rank IC (Spearman correlation between predictions and realised next-month returns) is ~0.04, which would classically be characterised as a weak signal. The Sharpe is nonetheless ~1.0 because **the strategy is not a high-conviction concentrated bet** — it holds ~440 positions per rebalance (k=20 × 11 GICS sectors, long + short), so the small per-name edge is multiplied by the law of large numbers. The Sharpe comes from broad cross-sectional structure (consistent small tilts toward many stocks) rather than from being right with high confidence on a few names.

Annualisation convention

All Sharpe ratios in this report use the `metrics.sharpe_ratio` convention — **geometric annualised return ÷ (monthly std, ddof=1, × √12)** — on the engine's stored `portfolio_returns` series (the realised, post-cost net return labelled by realisation date). On the corrected 14-feature pkl the choice of reduction barely matters: full-OOS, the canonical `sharpe_ratio` / `metrics.parquet` value is **+1.15**, the arithmetic mean/std × √12 is +1.14, and the population-std (ddof=0) variant is +1.14 — a spread under 0.02. We report **+1.15 full-OOS / +0.97 long-OOS / +1.00 test** (the §5 table) as the headline; the alternative reductions agree to within rounding and are noted here only for transparency.

Other limitations

- **Free-data coverage gap (pre-broader-rebuild).** ~10-16% of historical tickers (delisted/renamed before ~2022) are unavailable in yfinance under their old symbol; their history ends at the CRSP cutoff. The broader-universe rebuild via Sharadar closes this gap.
- **One ~12-year OOS sample.** The DSR adjustment accounts for the trials we ran, but it remains a single historical path.
- **Regime detection is far weaker out-of-sample than in-sample.** Fit on the full sample, the 2-state HMM flags **91.7%** of known stress months as crisis (macro-average over the seven episodes, GFC → 2022 inflation). Re-estimated honestly walk-forward (60-month burn-in, scaler refit at each step), that falls to **51.1%** — the very number used to select the model. The loss is concentrated in the short, fast wobbles: walk-forward, the HMM misses Euro I (0/4 months) and Q4-2018 (0/3) entirely, while still catching the longer episodes (China/Oil 100%, 2022 inflation 90%, COVID 67%); month-weighted (pooled) OOS recall is 64.5% (20/31 stress months). This is the same monthly-frequency timing limit §4 documents for the COVID crash, now quantified — the overlay reacts a rebalance late to brief shocks, which is also why it adds no drawdown protection on the broad book.

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7. Conclusion

A disciplined, survivorship-free, look-ahead-controlled ML pipeline produces a full-OOS net Sharpe of **+1.15** with a Fama-French 5-factor alpha of **+18.7%/yr at t = +6.85** over 2012-2024 on a broad survivorship-free universe (~4,400 names/month median, the alive set of the rolling top-2,000 US common stocks by market cap). The alpha survives Carhart momentum control (+20.1%/yr at t = +7.4, UMD $\beta = -0.43$, momentum-averse), block-bootstrap robustness ($P(SR \leq 0) = 0.0003$ long-OOS), conservative transaction-cost stress (significant up to ~50 bps/side), and a Deflated Sharpe Ratio of 0.87 at N = 25 trials. These checks span the principal referee questions a sceptical reader would raise — is the apparent edge just momentum?, is it just a lucky path?, is it priced out by costs?, was it cherry-picked across configurations? — and the answer in each case is no. The result is consistent with Gu, Kelly & Xiu (2020): cross-sectional ML alpha does exist on a broad survivorship-free US common-stock universe once both look-ahead and survivorship are correctly handled.

Three honest counterweights bound the claim. First, the strategy is **not market-neutral**: realised Mkt-β ≈ +1.3 and the long leg generates almost all of the P&L (+38%/yr vs −2%/yr for the short leg), so ~55% of the headline return is leveraged market exposure and the deepest drawdown is the −34% COVID-2020 crash that the monthly regime overlay cannot detect in time. Second, **the alpha is concentrated in the down-cap tail**: on the strict rolling top-2,000 (median ~2,000 names) the FF5 alpha collapses to +1.8%/yr (t = 0.96, n.s.) — consistent with GKX 2020 §IV.D but a binding limit on deployable AUM (capacity, realistic small-cap costs above 30 bps/side, and single-name fragility on the order of ±0.09 Sharpe per top contributor). Third, this study reports a single ~13-year historical OOS path on monthly data with 14 features; we did not test sub-monthly rebalancing, intraday execution, or post-2024 data. The defensible claim is therefore not "ML factor strategies work" — it is the narrower and more useful one: under realistic survivorship controls, point-in-time eligibility filters, and conservative costs, the Phase 24-RT canonical produces statistically significant cross-sectional alpha on the post-2015 OOS sample, and the path from the leaky pre-audit +1.49 Sharpe down to −0.31 (PIT-applied collapse on S&P-500-only) and back up to the honest +1.15 here is the methodological contribution that matters most.

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8. Reproducibility

All seeds pinned to 42. Data caches (gitignored) rebuild from a clean clone with `python -m notebooks.persona.run_all_data` (skips steps whose prerequisite — the CRSP raw CSV or the Nasdaq Data Link key — is absent). The canonical result and every diagnostic regenerate from the `notebooks/personb/*.py` phase scripts listed in the Alpha Model section §12; methodology figures from `python -m notebooks.persona.report_figures`. Every non-trivial design choice is logged in `DECISIONS.md`.

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Appendix B – Decision log (audit-era extract)

> Selected entries from `DECISIONS.md` documenting the Phase 11 – Phase 27b lineage and the survivorship-leak / Q-filter / INCLUDE_FEATURES audit. Full chronological log lives in `DECISIONS.md` in the repository.